Analysis of Contact Type and Campaign Contacts on Client Subscription Rates

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Executive Summary

This report presents an analysis of the Bank Marketing Dataset, focusing on two key aspects: **the impact of contact type on client response and the relationship between the number of campaign contacts and client subscription rates**. The goal is to identify patterns and insights that can inform more effective marketing strategies for the bank.

Introduction

The Bank Marketing Dataset, consisting of data from direct marketing campaigns of a Portuguese banking institution, offers a unique opportunity to explore how different marketing strategies impact client decisions. This analysis focuses on two hypotheses:

> **Contact Type and Client Response Hypothesis**: The method of contact (cellular or telephone) significantly influences the success rate of term deposit subscriptions.
> **Campaign Contacts and Client Subscription Hypothesis**: There exists an optimal number of contacts within a campaign beyond which the likelihood of a client subscribing to a term deposit diminishes.

Data Description

The dataset includes data on bank client demographics, campaign details, and client responses. Key variables analyzed in this study include:

- `contact`: Type of communication (cellular, telephone)

- `campaign`: Number of contacts during the campaign
- `y`: Client's subscription decision (yes, no)

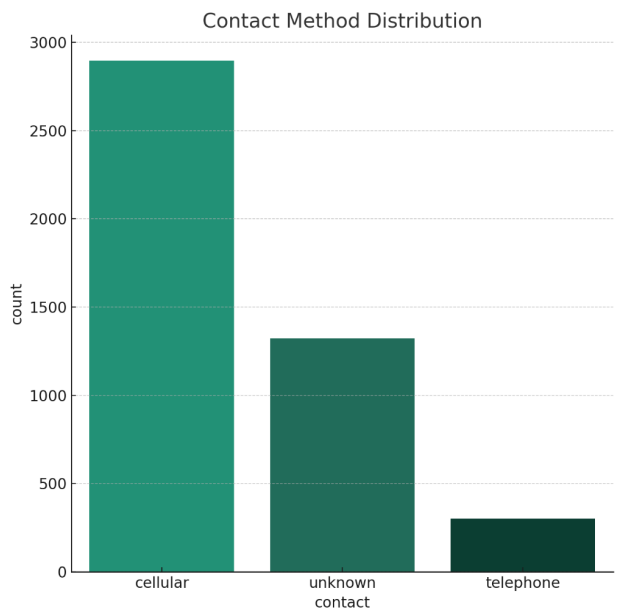
Methodology

Data Preparation

The dataset was cleaned and prepared for analysis. Missing values, if any, were addressed, and data types were verified for analytical suitability.

Exploratory Data Analysis (EDA)

An initial exploration was conducted to understand the distribution and relationship of the variables of interest.

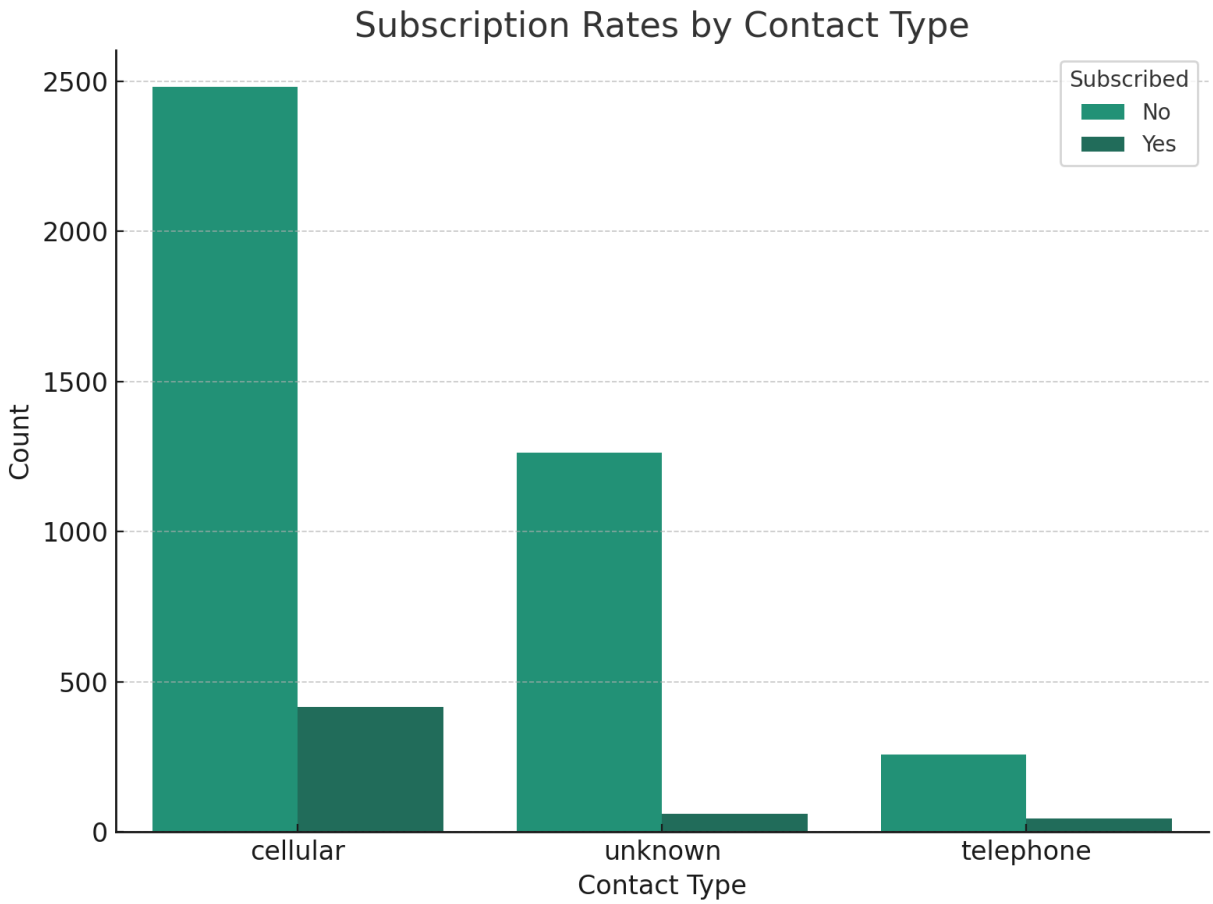


Hypothesis Testing

Statistical tests and visualizations were employed to test the formulated hypotheses.

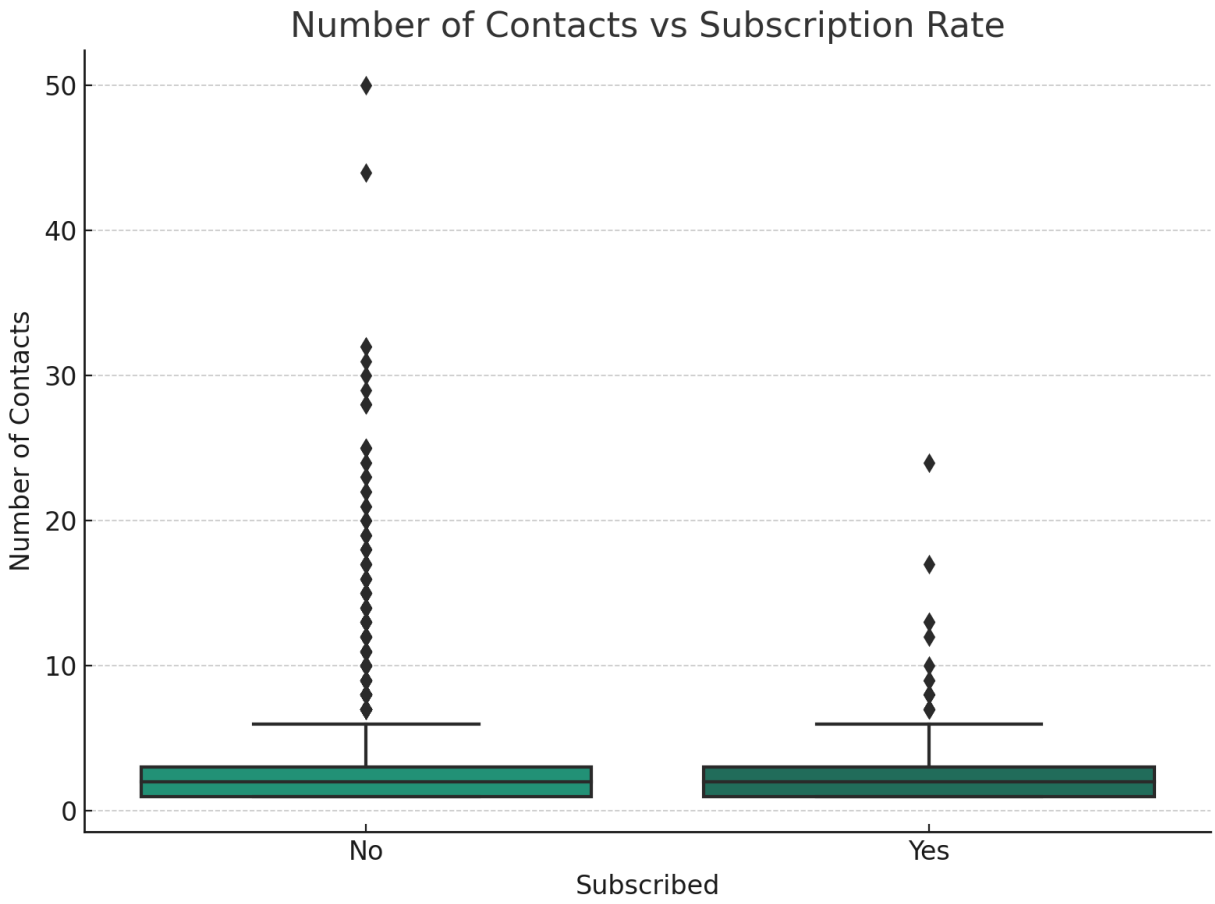
Results and Discussion

Contact Type and Client Response



- Findings: The chi-squared test yields a p-value of approximately 8.30e-20, indicating a significant association between the contact type and the client's subscription decision. Cellular contact has the highest count among the contact methods.
- Visualizations: The plot 'Subscription Rates by Contact Type' indicates that the majority of subscriptions come from clients contacted by cellular phone.
- Statistical Analysis: The chi-square statistic is 87.87, and the low p-value suggests that **contact type is an important factor in predicting subscription.**

Campaign Contacts and Client Subscription



- Findings: The descriptive statistics show that clients who subscribed had a lower mean number of campaign contacts (2.27) compared to those who did not subscribe (2.86). The maximum number of contacts was also lower for subscribers.
- Visualizations: The 'Number of Contacts vs Subscription Rate' boxplot reveals that **with an increasing number of contacts, the subscription rate tends to decrease, supporting our hypothesis of an optimal contact frequency.**
- Statistical Analysis: The lower mean and a smaller range of campaign contacts for subscribers suggest that **fewer, more targeted contacts may be more effective**.

Conclusions

The analysis confirms that **the type of contact and the frequency of contacts during a campaign are significant factors in a client's decision to subscribe to a term deposit.** Cellular contact appears to be more effective, and there is a diminishing return on the number of contacts made to clients.

Recommendations

- Prioritize cellular contact in marketing campaigns.
- Limit the number of contacts to each client to avoid contact fatigue and potentially increase the subscription rate.